Job Offer (Engineer / Data Scientist): Data Mapping & Machine Learning for Signature Integration – iRECORDS Project

- **Location:** Université de Versailles Saint-Quentin-en-Yvelines (UVSQ), Paris-Saclay, France – DAVID Laboratory
- **International Collaboration:** Interdisciplinary consortium including partners from Jena, Germany
- **Start Date:** As soon as possible
- **Duration:** 6 months
- **Supervisor:** Prof Zaineb Garcia, University of Lille

Project Context

Sepsis represents a major health, economic, and societal burden worldwide. In these patients, a dysregulated host immune response to infection is a hallmark of disease severity and supports the routine use of corticosteroids (CS), a low-cost and highly effective class of immunomodulatory drugs. However, patients respond heterogeneously to corticosteroid treatment. Stratifying patients according to their individual immune response profiles could significantly improve the benefit–risk balance of corticosteroid therapy. The iRECORDS ERA PERMED funded project (https://www.fhu-sepsis.uvsq.fr/i-records-3) aims to generate predictive signatures of corticosteroid sensitivity or resistance in patients with sepsis.

Objectives

The position focuses on developing data-driven methodologies to construct robust predictive signatures from multi-cohort sepsis datasets within the iRECORDS consortium.

1) Data Mapping & Harmonization
    - Map heterogeneous datasets from different cohorts.
    - Identify correspondences between biological variables across studies.
    - Ensure semantic and structural consistency across datasets.
    - Build an integrated, multi-cohort dataset suitable for machine learning analyses.

2) Machine Learning Pipeline Development
    - Design and implement a full data mining pipeline, including pre-processing, feature engineering, and handling missing data.
    - Develop and benchmark predictive ML models.
    - Compare and evaluate multiple ML approaches.
    - Assess model robustness, interpretability, and generalizability.

Goal: Produce a reliable predictive signature to identify patient responsiveness to corticosteroid therapy.

The selected candidate will work in close collaboration with an interdisciplinary and international team across France and Germany, interacting with clinicians, biologists, and computational scientists.

Key Responsibilities

- Integrate and harmonize multi-cohort datasets.
- Develop, evaluate, and document machine learning workflows.
- Contribute to consortium-wide methodological discussions.
- Ensure reproducibility and transparency of all analyses.
- Present findings to an international, interdisciplinary team of clinicians, biologists, and computational scientists.

Candidate Profile

We are looking for a highly motivated candidate with:

- Background in Computer Science, Data Science, Bioinformatics, or related fields.
- Strong knowledge of machine learning and predictive modeling.
- Proficiency in Python and related data science libraries.
- Experience with data preprocessing, feature engineering, and modeling workflows.
- Interest in biomedical research and translational applications.
- Ability to work independently and collaboratively in interdisciplinary teams.
- Strong communication skills in English.

Application Instructions:

Please send your application to zaineb.garcia@univ-lille.fr including:

- Academic transcripts
- Motivation letter
- Contact details of two referees

Subject line: “iRECORDS application”